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Expanded Program of Immunization

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IMMUNIZATION (Vaccination)



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Immunity

Immunity is the ability of the human body to tolerate the presence of material ^(native) indigenous to the body (“self”), and to eliminate foreign (“non self”) material.

Classification of Vaccine



- 1- Live attenuated** (live weakened) ex: oral Polio vaccine (OPV), ROTA, yellow fever, measles, mumps, rubella, BCG (Bacillus-Calmette-Guérin)
- viral
 - bacterial

2- Inactivated Inactivated Vaccines (killed)

- Whole (ex: whole cell Pertussis, inactivated Polio)
 - Viruses
 - bacteria
- Fractional
 - protein-based
 - toxoid (ex: tetanus, diphtheria)
 - subunit (ex: HBV, Acetabular Pertussis)
 - polysaccharide-based (ex: pneumococcal, meningococcal)
 - pure
 - conjugate (Weak antigen + strong carrier) (ex: pneumococcal, Haemophilus influenza B)

Note: ((the more similar the vaccine is to the disease-causing form of the organism, the better the immune response to the vaccine))



Live Attenuated Vaccine

Advantages:

- Attenuated (weakened) form of the “wild” virus or bacterium
- Must replicate to produce an immune response
- Immune response virtually identical to natural infection → very strong immune response (stronger than inactivated)
- Usually produce immunity with one dose

Disadvantages:

- Severe reactions possible → may cause serious infection
- Interference from circulating antibody
- Fragile - must be stored and handled carefully

Viral: measles, mumps, rubella,, varicella, zoster, yellow fever, rotavirus, intranasal influenza, oral polio

Bacterial: BCG, oral typhoid

Inactivated Vaccines



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Advantages:

- Cannot replicate *So severe reaction highly unlikely*
- Less affected by circulating antibody than live vaccines

Disadvantages

Useful for immunocompromised patients who can't get live-attenuated

- Always require multiple doses
- Immune response mostly humoral *Whilst live attenuated vaccine is cellular + humoral*
- Antibody titer diminish with time
- May require periodic supplemental booster doses

☐ Whole-cell vaccines

- viral: polio, hepatitis A, rabies
- bacterial: pertussis*, typhoid*, cholera*, plague*

☐ Fractional vaccines

☐ Subunits: hepatitis B, influenza, acellular pertussis, human papillomavirus, anthrax

☐ Toxoids: diphtheria, tetanus



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Expanded Program of Immunization



Scope of the EPI

Experience with smallpox eradication program showed the world that immunization was the most powerful and cost-effective weapon against vaccine preventable diseases.

In 1974, the WHO launched its “ Expanded program of immunization” (EPI) against six most common preventable diseases (diphtheria, pertussis, tetanus, polio, tuberculosis and measles.



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“**Expanded**” means:

- Expanding the number of diseases to be covered
- Expanding the number of children and target population to be covered
- Expanding coverage to all corners of the country and spreading services to reach the less privileged sectors of the society



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Targets

- **Under 5-years children.**
- **Women in the child bearing age (15-49 years).**
- **Special Groups Vaccination**



Vaccination schedule in IRAQ (current)

→ EPI under 5

ign 20

Age	Type of vaccine
0-1 Week	OPV0 , HepB1 , BCG <i>although it's live-attenuated it needs both</i> <i>→ it's vaccines of 5 pathogens combined: Tetanus, Pertussis, Diphtheria, Hep B, Haemophilus-influenza-B</i>
2 months	OPV1 , PENTA, ROTA1* + PCV1 <i>multiple dose and booster dose</i> <i>→ pneumococcal conjugate vaccine</i>
4 months	OPV2 , PENTA, ROTA2* , IPV1+PCV2 <i>→ inactivated Polio vaccine</i>
6 months	OPV3 , PENTA, IPV2+ PCV3 + Rota 3*
9 months	Measles1 + VIT A (100,000 IU)
12 months	MMR1 (Measles , Mumps , Rubella)
18 months	DPT, OPV 1 st Booster dose VIT A (200,000 IU) + MMR2 <i>diphtheria - pertussis - tetanus</i>
4-6 years	DPT , OPV 2 nd Booster dose Note: Penta (D.wP.T. + HBV + Hib)



Rota Vaccines

- Rota teq (Penta Valent)

↳ number of different serotypes (strains of a specific M.O. the vaccine target (protective against))

- Rota Rix (Mono Valent)

same letter; mono 1st time beginning

- ROTASIIL-Liquid (Penta Valent)



→ Tetanus, Toroid.

→ Child Bearing Age

T.T. Schedule in CBA (15- 49 Y)

1st dose

(no protection)

→ usually at 4th month of pregnancy it could be done even if not pregnant and only need to be done once
the whole 5 dose

2 important notes

① if you do it during pregnancy you shall need do it in 1st trimester.

2nd dose after one month (6 M- 3 Y)

② if you do like 2 doses and forget about it for 5 years you start from beginning

3rd dose after six months (5Y)

at 2nd dose

4th dose after one year (10 Y)

5th dose after one year (life long)



Adverse events following vaccination

vaccine	mild	sever
BCG	Axillary, cervical, Lymph adenitis	Osteitis TB meningitis
Diphtheria	Redness, pain, fever	Urticaria, pruritis anaphylaxis
Tetanus	Pain, erythema sterile-abscess, fever	Urticaria, brachial neuritis Gullian-Barre anaphylaxis
Pertussis	Pain, tenderness, Nodule erythema, oedema fever, irritation, loss appetite, vomiting	* persistent inconsolable cry * unusual screams * convulsion * fever 40.5° c * hypotonic- hyporesponsive episodes * encephalopathy
Hepatitis B	Fever, pain, swelling erythema, headache	Anaphylaxis Gullian-Barre Hair-loss Diabetes type 1
Measles	Pain, tenderness, fever , rash	Urinary, encephalopathy, encephalitis, SSPE, seizures Gullian-Barre thrombocytopenia, Autism inflammatory bowel disease
Mumps	Pain, tenderness, fever parotitis	Aseptic meningitis
Rubella	Pain, tenderness, fever rash, lymphadenopathy headache	Arthralgia Arthritis Arthropathy
MMR	Fever, rash, parotitis lymphadenopathy	
OPV		VAPP, Gullian-Barre aseptic meningitis, transverse myelitis

Haemophilus influenza type-b- vaccines

- Hib polysaccharide conjugated vaccine
- Available in 1987

T-cell dependent—Effective^{even} below 18 months of age

Antibodies are IgG—Reduce carrier state

Repeated doses elicit booster response

Vaccination schedule

All infants ,including those born prematurely, should receive a primary series ^{in PENTA vaccine} of conjugate Hib vaccine

- Primary series is 3 doses 2,4,6 months of age
- Booster dose is recommended
- The recommended interval between primary series doses is 8 weeks with a minimum interval is 4 weeks
- Hib vaccine should never be given to a child younger than 6 weeks of age

➤ Use in older children and adults

- Generally not recommended for persons older than 59 months of age ^{5 years}
- Consider for high risk persons e.g.

➤ Asplenia

➤ Immune-deficiency

➤ HIV

➤ HSCT (haemopoietic stem cell transplant)



Hib vaccine precaution and contraindication

- Anaphylactic reaction to previous dose or to one of the vaccine component
- Moderate or severe acute illness _____

• Age younger than 6 weeks

→ similar to RTS₂ vaccine in malaria R21

26. High-risk adults needing Hib vaccine include:

- A. Asplenic and immunocompromised patients
- B. Healthy adults
- C. All over 60 years
- D. None – only children need it



Hib vaccine adverse reactions

- Swelling ,redness, pain in 5-30% of recipients
- Systemic reactions infrequent
- Serious adverse reaction is rare



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Pneumococcal Disease

(before introducing pneumococcal vaccine into the schedule of Vaccination)

→ Missing
controlled
Pneumococcal is now fully

- Most common cause of vaccine- preventable death in the U.S.
- Most common cause of bacterial meningitis among infants and young children
- Increasing antibiotic resistance

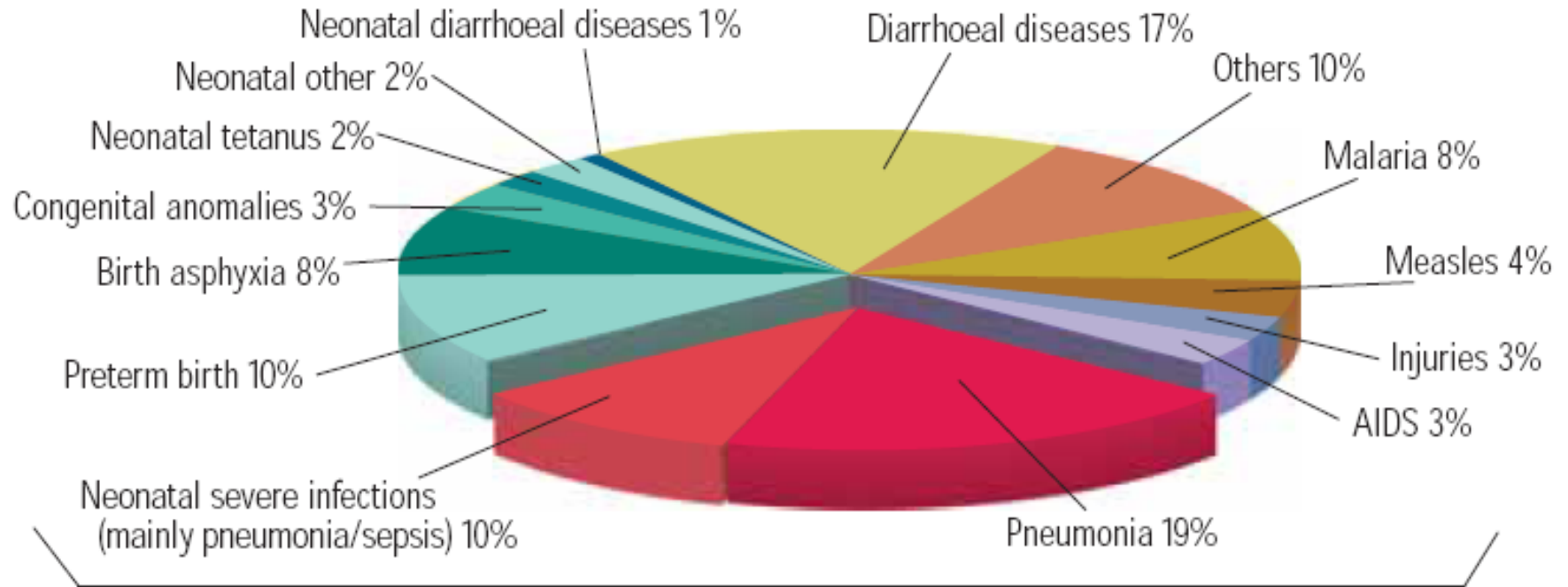
Pneumococcal Disease Clinical Syndromes

- Pneumonia
- Bacteremia
- Meningitis

Pneumonia is the Leading Killer of Children Worldwide

JK

Global distribution of cause-specific mortality among children under five, 2004



Pneumococcal Vaccines

- 1977 14-valent polysaccharide vaccine licensed
- 1983 23-valent polysaccharide vaccine licensed
- 2000 7-valent polysaccharide conjugate vaccine licensed

Pneumococcal Polysaccharide Vaccine

Recommendations *not the converse*

- Adults ≥ 65 years of age
- Persons ≥ 2 years with
 - chronic illness
 - anatomic or functional asplenia
 - immunocompromised (disease, chemotherapy, steroids)
 - HIV infection
 - environments or settings with increased risk



Pneumococcal Conjugate Vaccine

- Routine vaccination of children age <24 months and children 24-59 months with high risk medical conditions
- Doses at 2, 4, 6, months, booster dose at 12-15 months
Unvaccinated children >7 months require fewer doses



Pneumococcal Polysaccharide Vaccine Revaccination

- Routine revaccination of immuno-competent persons is not recommended
- Revaccination recommended for persons age ≥ 2 years at highest risk of serious pneumococcal infection
- Single revaccination dose ≥ 5 years after first dose

Pneumococcal Vaccines Contraindications and Precautions



- Severe allergy to vaccine component or following prior dose of vaccine
- Moderate to severe acute illness



Current Pneumococcal Conjugate Vaccines

PCV 7	Carrier: CRM ₁₉₇	4	6B	9V	14	18C	19F	23F	52					
PCV 10	Carriers: □ Protein D □ Tetanus toxoid □ Diphtheria Toxoid	4	6B	9V	14	<u>18C</u>	<u>19F</u>	23F	1	5	7F			
PCV 13	Carrier: CRM ₁₉₇	4	6B	9V	14	18C	19F	23F	1	5	7F	3	6A	19A



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Influenza Vaccine



Types Of Influenza vaccine

→ need to be renewed yearly

- Seasonal Inactivated Influenza Vaccines (TIV or QIV)

Tri-valent - Influenza - Vaccine

↳ Quadri (tetra) - valent - influenza vaccine

- Live- Attenuated Influenza Vaccine (LAIV)

Vaccine Effectiveness

- ❑ Influenza immunization prevents disease in 52-82% of healthy individuals
- ❑ In the elderly vaccine effectiveness is about half of that of healthy adults;
- ❑ however influenza immunization decreases the incidence of Pneumonia, Hospital admission and Death in the elderly, and reduces exacerbations in persons with COPD

People who have Priority in receiving Influenza Vaccine

- Children < 5, but especially children younger than 2 years old
- Adults 65 years of age and older
- Pregnant women (and women up to two weeks postpartum)
- Residents of nursing homes and other long-term care facilities
- Health Practitioner (HCW)
- Immunizers



People who have medical conditions

- Neurological : (cerebral palsy, epilepsy , stroke, intellectual disability, developmental delay, muscular dystrophy, or spinal cord injury)
- [Asthma](#) & Chronic lung disease: COPD and cystic fibrosis)
- [Heart disease](#) (such as congenital heart disease, congestive HF & CAD)
- Blood disorders (such as sickle cell disease)
- Endocrine disorders (such as [diabetes](#))
- Kidney & Liver disorders
- Metabolic disorders (such as inherited metabolic disorders and mitochondrial disorders)
- Immunodeficiency : ([HIV or AIDS](#), or [cancer](#), or those on chronic steroids)
- People <19 years of age who are receiving long-term aspirin therapy
- People with extreme obesity: BMI of 40 or more)



Regarding the Vaccine

- 0.5 cc
- Single Dose: Pre-filled, single dose syringe
- 6 months of age and older

never in younger than 6 month

Influenza vaccine dosing for specific ages

➤ 6 months up to & including 8 years of age

- 2 doses* if never previously immunized with seasonal influenza vaccine (spaced 4 weeks apart –minimum interval)
- 1 dose only if previously immunized with seasonal influenza vaccine

➤ 9 years of age and older : 1 dose *without seasonal influenza vaccine*

❑ From 6 M upto 35 M: (0.25 cc) While From 36 M and older (0.5) cc



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Common Reactions (AEFA)

- Injection site redness, swelling, pain
- Fatigue, headache, myalgia
- Arthralgia, fever, chills, malaise



Rare Reactions

- Immediate, allergic-type responses such as hives,
angioedema, allergic asthma, systemic anaphylaxis
- Guillain-Barré Syndrome (GBS)
- Oculorespiratory Syndrome (ORS)

→ Wierafid

Contraindications to TIV → Influenza Vaccine

- Age: less than 6 months of age
- Have had an anaphylactic reaction to a previous dose of influenza vaccine
- Have a known hypersensitivity to any component of the vaccine with the

Exception of Egg

- Have been diagnosed with Guillain-Barré Syndrome within 6 weeks of a previous dose of influenza vaccine
- Have experienced **severe Oculorespiratory Syndrome** within **24 hrs** of **receiving influenza immunization**

Egg-allergic individuals

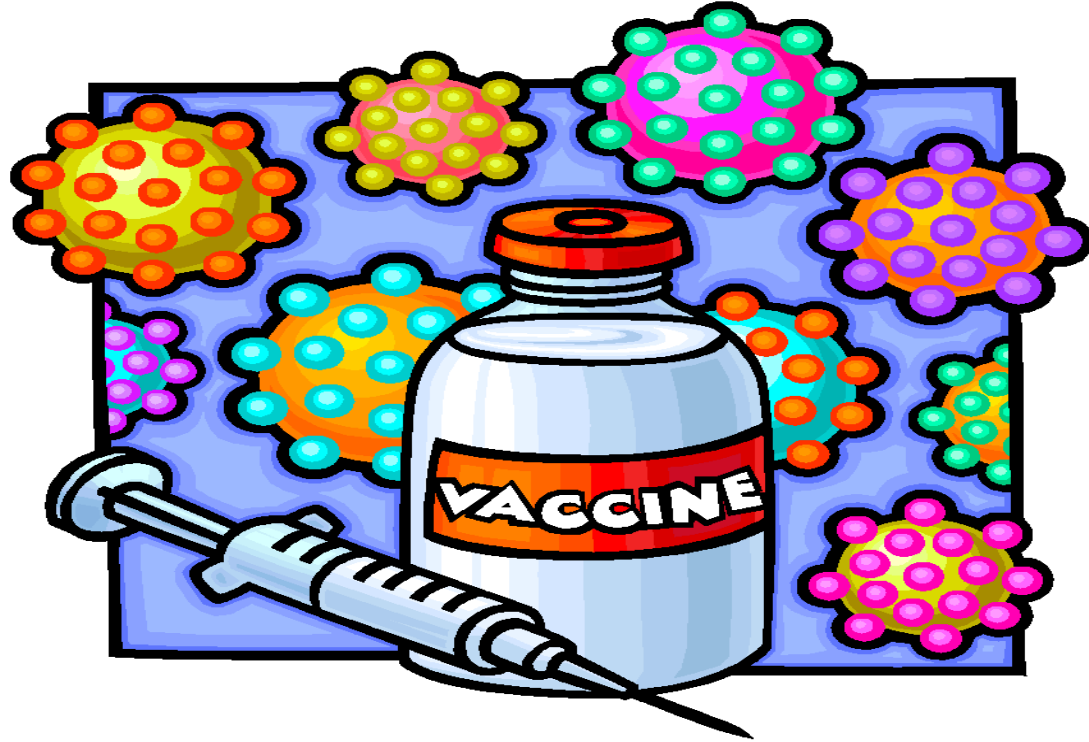
- Egg allergy is no longer considered a contraindication for influenza vaccine
- Egg-allergic individuals may be immunized without a prior influenza vaccine skin test and with the full dose of vaccine, irrespective of a past severe reaction to egg

Vaccine deferral

- Those with **serious acute febrile illness usually should not be immunized until symptoms have abated**
- **Vaccine does not require deferral and can safely be given to the following individuals:**
 - A. Those with mild acute illness, with or without fever
 - B. Individuals who are recovering from illness or are taking antibiotics



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Thank you

A. Basics of Immunity and Vaccine Types

1. Immunity is defined as:

- A. The ability of the body to eliminate self-antigens
- B. The ability to tolerate self and eliminate foreign substances
- C. The destruction of self-cells by the immune system
- D. Tolerance to all external agents

Answer:

 B. The ability to tolerate self and eliminate foreign substances

2. Which of the following vaccines contains a living organism?

- A. Tetanus toxoid
- B. Hepatitis B
- C. BCG
- D. Influenza inactivated vaccine

Answer:

 C. BCG

3. Live attenuated vaccines are preferred because:

- A. They require multiple doses
- B. They produce immunity identical to natural infection
- C. They cannot replicate in the host
- D. They are not affected by antibodies

Answer:

☒ B. They produce immunity identical to natural infection

4. The major disadvantage of live attenuated vaccines is:

- A. Weak immune response
- B. High cost
- C. Severe reaction in immunocompromised persons
- D. No long-term protection

Answer:

☒ C. Severe reaction in immunocompromised persons

5. Which of the following is an inactivated whole viral vaccine?

- A. OPV
- B. IPV
- C. BCG
- D. Measles

Answer:

☒ B. IPV

6. "The more similar the vaccine is to the disease-causing organism, the better the immune response" – this statement best explains:

- A. The effect of preservatives
- B. The basis of live attenuated vaccines
- C. The principle of toxoid vaccines
- D. Antibody half-life

Answer:

☒ B. The basis of live attenuated vaccines

7. Which of the following vaccines is a toxoid?

- A. Measles
- B. Tetanus
- C. Polio
- D. Influenza

Answer:

 B. Tetanus

8. Which of the following vaccines is fractional and subunit?

- A. BCG
- B. Hepatitis B
- C. DTP
- D. OPV

Answer:

 B. Hepatitis B

 B. Expanded Program on Immunization (EPI)

9. The Expanded Program on Immunization (EPI) was launched by WHO in:

- A. 1968
- B. 1974
- C. 1983
- D. 1990

Answer:

 B. 1974

10. The EPI originally targeted which six diseases?

- A. Diphtheria, Pertussis, Tetanus, Polio, TB, Measles
- B. Tetanus, Hepatitis, Typhoid, Meningitis, Polio, TB
- C. Diphtheria, Hepatitis, Measles, Rubella, Polio, TB
- D. Tetanus, Typhoid, Mumps, Rubella, Polio, TB

Answer:

 A. Diphtheria, Pertussis, Tetanus, Polio, TB, Measles

11. "Expanded" in EPI refers to:

- A. Including new vaccines, more children, and wider coverage
- B. Introducing boosters only
- C. Expanding adult vaccination only
- D. Exporting vaccines to other countries

Answer:

 A. Including new vaccines, more children, and wider coverage

12. Target groups in EPI include:

- A. Only newborns
- B. Under-five children and women of childbearing age
- C. Only pregnant women
- D. Only high-risk adults

Answer:

 B. Under-five children and women of childbearing age

13. At birth, infants in Iraq receive which vaccines?

- A. BCG + OPV0 + HepB1
- B. DTP + IPV + Rota
- C. OPV1 + Measles
- D. DTP + MMR

Answer:

➡ A. BCG + OPV0 + HepB1

14. The "Penta" vaccine contains:

- A. DTP + Hib + Hepatitis B
- B. DTP + Measles + Polio
- C. Hepatitis A + Hib + Influenza
- D. DTP + Polio + Rotavirus

Answer:

➡ A. DTP + Hib + Hepatitis B

15. The first measles vaccine is given at:

- A. 6 months
- B. 9 months
- C. 12 months
- D. 18 months

Answer:

☒ B. 9 months

16. The MMR vaccine protects against:

- A. Measles, Malaria, Rubella
- B. Mumps, Measles, Rubella
- C. Measles, Mumps, Rotavirus
- D. Mumps, Measles, Hepatitis

Answer:

☒ B. Mumps, Measles, Rubella

17. Vitamin A supplementation is given at which ages in the EPI schedule?

- A. 2 and 6 months
- B. 9 and 18 months
- C. 12 and 24 months
- D. 6 weeks and 9 months

Answer:

☒ B. 9 and 18 months

18. What is the correct Tetanus Toxoid schedule for women of childbearing age?

- A. 3 doses 6 months apart
- B. 5 doses over several years for lifelong protection
- C. 2 doses only
- D. Annual booster forever

Answer:

☒ B. 5 doses over several years for lifelong protection

⚠ D. Vaccine Adverse Events

19. Persistent inconsolable cry after vaccination is a reaction to:

- A. Hepatitis B
- B. Pertussis
- C. Measles
- D. BCG

Answer:

➡ B. Pertussis

20. Which vaccine may rarely cause Vaccine-Associated Paralytic Polio (VAPP)?

- A. IPV
- B. OPV
- C. DTP
- D. Hib

Answer:

➡ B. OPV

21. Measles vaccine may rarely cause:

- A. Guillain-Barré syndrome and SSPE
- B. Meningitis
- C. Diabetes
- D. Hair loss

Answer:

 A. Guillain-Barré syndrome and SSPE

22. The most serious contraindication to vaccination is:

- A. Mild fever
- B. History of anaphylaxis to a previous dose
- C. Cough and cold
- D. Use of antibiotics

Answer:

 B. History of anaphylaxis to a previous dose

 E. Haemophilus Influenzae Type b (Hib) Vaccine

23. Hib vaccine is:

- A. Live attenuated
- B. Inactivated whole bacterial
- C. Polysaccharide conjugate
- D. Viral subunit

Answer:

 C. Polysaccharide conjugate

24. Primary Hib vaccination schedule:

- A. 2, 4, 6 months + booster
- B. 1 dose at birth
- C. 3 doses in 1st month

D. Only after 1 year

Answer:

➡ A. 2, 4, 6 months + booster

25. Hib vaccine is not given before:

A. 2 months

B. 6 weeks

C. 3 months

D. 1 year

Answer:

➡ B. 6 weeks

26. High-risk adults needing Hib vaccine include:

A. Asplenic and immunocompromised patients

B. Healthy adults

C. All over 60 years

D. None – only children need it

Answer:

➡ A. Asplenic and immunocompromised patients



F. Pneumococcal Vaccines

27. Pneumococcal disease is a leading cause of:

- A. Viral meningitis
- B. Bacterial meningitis and pneumonia
- C. Tetanus
- D. Diarrhea

Answer:

 B. Bacterial meningitis and pneumonia

28. Routine pneumococcal vaccination starts at:

- A. 6 months
- B. 2 months
- C. 9 months
- D. 1 year

Answer:

 B. 2 months

29. The pneumococcal conjugate vaccine (PCV) is recommended for:

- A. Children <2 years
- B. Adults only
- C. Pregnant women only
- D. Not in EPI schedule

Answer:

 A. Children <2 years

30. PCV13 covers how many serotypes?

- A. 7
- B. 10
- C. 13
- D. 23

Answer:

 C. 13

 G. Influenza Vaccine

31. The live-attenuated influenza vaccine is given:

- A. By injection
- B. Orally
- C. Intranasally
- D. Subcutaneously

Answer:

 C. Intranasally

32. The influenza vaccine should be given annually because:

- A. Immunity is lifelong
- B. Virus strains change every year
- C. Boosters are dangerous
- D. It causes infertility

Answer:

 B. Virus strains change every year

33. Children aged 6 months to 8 years receiving flu vaccine for the first time should get:

- A. 1 dose

- B. 2 doses 4 weeks apart
- C. 3 doses
- D. Half-dose only

Answer:

 B. 2 doses 4 weeks apart

34. Common adverse reaction of influenza vaccine:

- A. Redness, mild fever, myalgia
- B. Severe paralysis
- C. Rash and jaundice
- D. Hypoglycemia

Answer:

 A. Redness, mild fever, myalgia

35. A rare complication of influenza vaccine:

- A. SSPE
- B. Guillain-Barré Syndrome
- C. Sepsis
- D. Hair loss

Answer:

 B. Guillain-Barré Syndrome

36. Influenza vaccination is contraindicated in:

- A. Mild cold
- B. Age below 6 months
- C. Pregnancy
- D. Diabetes

Answer:

 B. Age below 6 months

 H. Special Notes & General Concepts

37. Which of the following vaccines is safe and recommended in pregnancy?

- A. Live attenuated vaccines
- B. Influenza inactivated vaccine
- C. MMR
- D. Varicella

Answer:

 B. Influenza inactivated vaccine

38. Which of the following is a conjugate vaccine?

- A. Tetanus toxoid
- B. PCV
- C. OPV
- D. DTP

Answer:

 B. PCV

39. Severe acute febrile illness is a reason to:

- A. Delay vaccination until recovery
- B. Avoid all future vaccines

- C. Proceed with vaccination
- D. Use half dose

Answer:

☒ A. Delay vaccination until recovery

40. Which statement about egg allergy and flu vaccine is correct?

- A. Egg allergy is a contraindication
- B. Skin testing is required before vaccination
- C. Egg allergy no longer prevents flu vaccination
- D. Vaccine must be avoided completely

Answer:

☒ C. Egg allergy no longer prevents flu vaccination
